Course notes

GCD and Euclid's algorithm

Definition · GCD. Let a and b be two integers, not both zero. The set of their positive common divisors is finite and contains 1; its largest element is called the *greatest common divisor* of a and b, written $\gcd(a,b)$. We say that a and b are *coprime* if $\gcd(a,b)=1$.

Property · Euclid's lemma. Let a and b be two integers with $b \neq 0$. For every integer k, the common divisors of a and b are those of b and $a-kb$. In particular, if r is the remainder of the division of a by $b \geq 1$, then $\gcd(a,b)=\gcd(b,r)$.

Proof. If $d \mid a$ and $d \mid b$, then $d \mid a-kb$. Conversely, if $d \mid b$ and $d \mid a-kb$, then $d \mid (a-kb)+kb=a$. The two sets of common divisors are equal, so their largest elements are too. □

Method · Euclid's algorithm. Replace (a,b) by (b,r) as long as the remainder r is not zero. The remainders form a strictly decreasing sequence of natural numbers, so the algorithm stops. The gcd is the last nonzero remainder. Example:

$$1{,}053 = 3 \times 325 + 78, \qquad 325 = 4 \times 78 + 13, \qquad 78 = 6 \times 13 + 0,$$

so $\gcd(1{,}053, 325) = 13$.

Bézout's theorem

Property · Writing the gcd. Let a and b be two integers, not both zero. There are integers u and v such that $au+bv=\gcd(a,b)$. Moreover, every common divisor of a and b divides $\gcd(a,b)$.

Proof. Let E be the set of positive integers of the form $au+bv$, with u, v integers. E is not empty (it contains a^2+b^2). Let $d=au_0+bv_0$ be its smallest element.

Divide a by d: $a=dq+r$ with $0 \leq r < d$. Then $r=a-q(au_0+bv_0)=a(1-qu_0)+b(-qv_0)$ is of the form $au+bv$. If it were positive, it would belong to E while being smaller than d: impossible. So $r=0$ and $d \mid a$. Likewise $d \mid b$.

Finally every common divisor c of a and b divides $au_0+bv_0=d$, so $c \leq d$. Hence d is the greatest common divisor: $d=\gcd(a,b)$. □

Property · Bézout's theorem. Two integers a and b are coprime if and only if there are integers u and v such that $au+bv=1$.

Proof. If $\gcd(a,b)=1$, the previous theorem provides u and v. Conversely, if $au+bv=1$, every positive common divisor of a and b divides 1, hence equals 1. □

Method · Bézout pair: working back through Euclid's algorithm. With the previous example: $13 = 325 - 4 \times 78 = 325 - 4(1{,}053 - 3 \times 325) = 13 \times 325 - 4 \times 1{,}053$. A suitable pair for $(1{,}053, 325)$ is $(-4, 13)$.

Key point · Extended Euclidean algorithm in Python. The following function returns (d,u,v) with $d=\gcd(a,b)=au+bv$, for natural numbers a and b not both zero.

```
def euclide_etendu(a, b):
    u0, v0, u1, v1 = 1, 0, 0, 1
    while b != 0:
        q, r = a // b, a % b
        a, b = b, r
```

```
        u0, u1 = u1, u0 - q * u1
        v0, v1 = v1, v0 - q * v1
    return a, u0, v0
```

At every step, the two current remainders can be written $au_0 + bv_0$ and $au_1 + bv_1$ in terms of the initial values of a and b.

Gauss's theorem

Property · Gauss's theorem. Let a, b, c be three integers. If $a \mid bc$ and a and b are coprime, then $a \mid c$.

Proof. By Bézout's theorem, there are u and v such that $au + bv = 1$. Multiplying by c: $c = acu + bcv$. Now $a \mid acu$ and $a \mid bc$, so a divides $acu + bcv = c$. □

Property · Corollary. If a and b are coprime and both divide n, then ab divides n.

Proof. $n = ak$. Since $b \mid ak$ and $\gcd(a, b) = 1$, Gauss gives $b \mid k$: $k = bk'$ and $n = abk'$. □

Property · Inverse modulo n. Let $n \geq 2$ and a an integer. There is an integer u such that $au \equiv 1 \pmod{n}$ if and only if $\gcd(a, n) = 1$. Such a u is an *inverse of a modulo n*; it allows us to solve $ax \equiv b \pmod{n}$: $x \equiv ub \pmod{n}$.

Proof. $au \equiv 1 \pmod{n}$ is equivalent to the existence of an integer v such that $au + nv = 1$: this is Bézout's theorem. □

Method · Equation $ax + by = c$ in integers.

1. It has solutions if and only if $\gcd(a, b)$ divides c; we then divide by this gcd.
2. Find a particular solution (x_0, y_0).
3. By subtraction, $a(x - x_0) = -b(y - y_0)$; with a and b coprime, Gauss gives $x - x_0 = bk$, then $y - y_0 = -ak$.
4. Check that the pairs $(x_0 + bk, y_0 - ak)$, $k \in \mathbb{Z}$, are indeed solutions.

Example: $7x + 5y = 1$ has the particular solution $(-2, 3)$; its solutions are the pairs $(-2 + 5k, 3 - 7k)$, $k \in \mathbb{Z}$.